

Cesar Giner

Program 3 - AI Incorporated Code and LLM Assist

Notes:

- For this program, I mainly used LLM to assist me in conceptualizing ideas to design the implementation of the evaluator. The spec file from the course, 'CS 441 Spring 2026 Program 3: The Execution Engine (Interpreter)' was a pretty definitive guide to implement the final step of our interpreter. So, not much code was generated or used by the LLM, other than testing code, and code to fix a potential zero-div error.
- Link to LLM: <https://claude.ai/share/4194fef3-7c93-4491-901f-b3e07dedb8a1>
- Incorporated Code Below:

A) Testing

```
racket

; Basic assignment
(run "x := 5;")

; Assignment with arithmetic expression
(run "x := 1 + 2 * 3;")

; Print a variable
(run "x := 10; print x;")

; Print an expression directly
(run "print 2 + 3;")

; If statement - true branch
(run "x := 5; if (x > 3) { print x; }")

; If statement - false branch
(run "x := 1; if (x > 3) { print x; }")

; Variable reassignment (tests environment threading)
(run "x := 5; x := 10; print x;")

; While loop
(run "x := 0; while (x < 3) { x := x + 1; } print x;")

; Nested arithmetic
(run "x := (2 + 3) * (4 - 1); print x;")

; Undefined variable lookup (should trigger error)
(run "print y;")
```

B) Bug Fix

```
Check the exact eval-expr division line >
Replace line 384 with:

racket

[[equal? op '/') (if (= right 0)
  (error "Runtime Error: division by zero")
  (/ left right))]
```